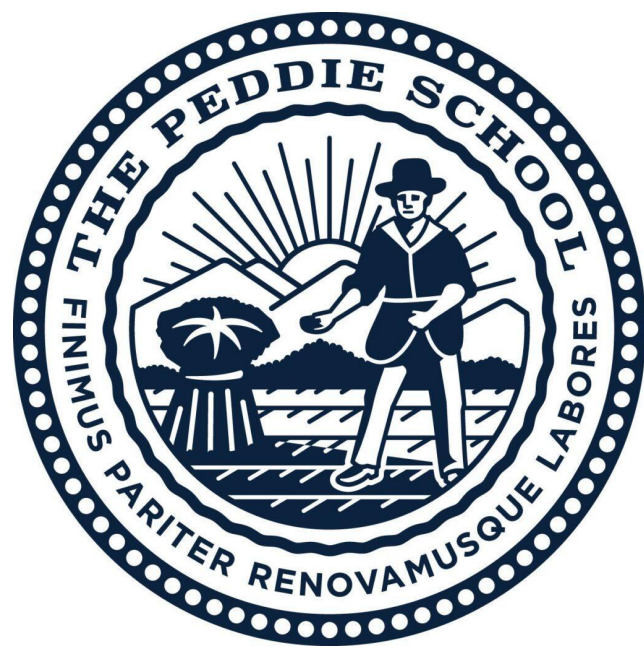




Analyzing OOD in VLMs: Can Confidence Detect Failure in Visual LLMs?

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Motivation

LLMs and VLMs are prone to hallucinations and are often overconfident, needing methods to detect failure.

Background

Out-of-Distribution (OOD) occurs when a model encounters testing data that is different from the training data (Figure 1, 2, 3).

For narrow classification models, logit-based OOD detection methods like MSP, Entropy, Max Logit, and Energy are able to detect unusual inputs and model failures (Figure 4).

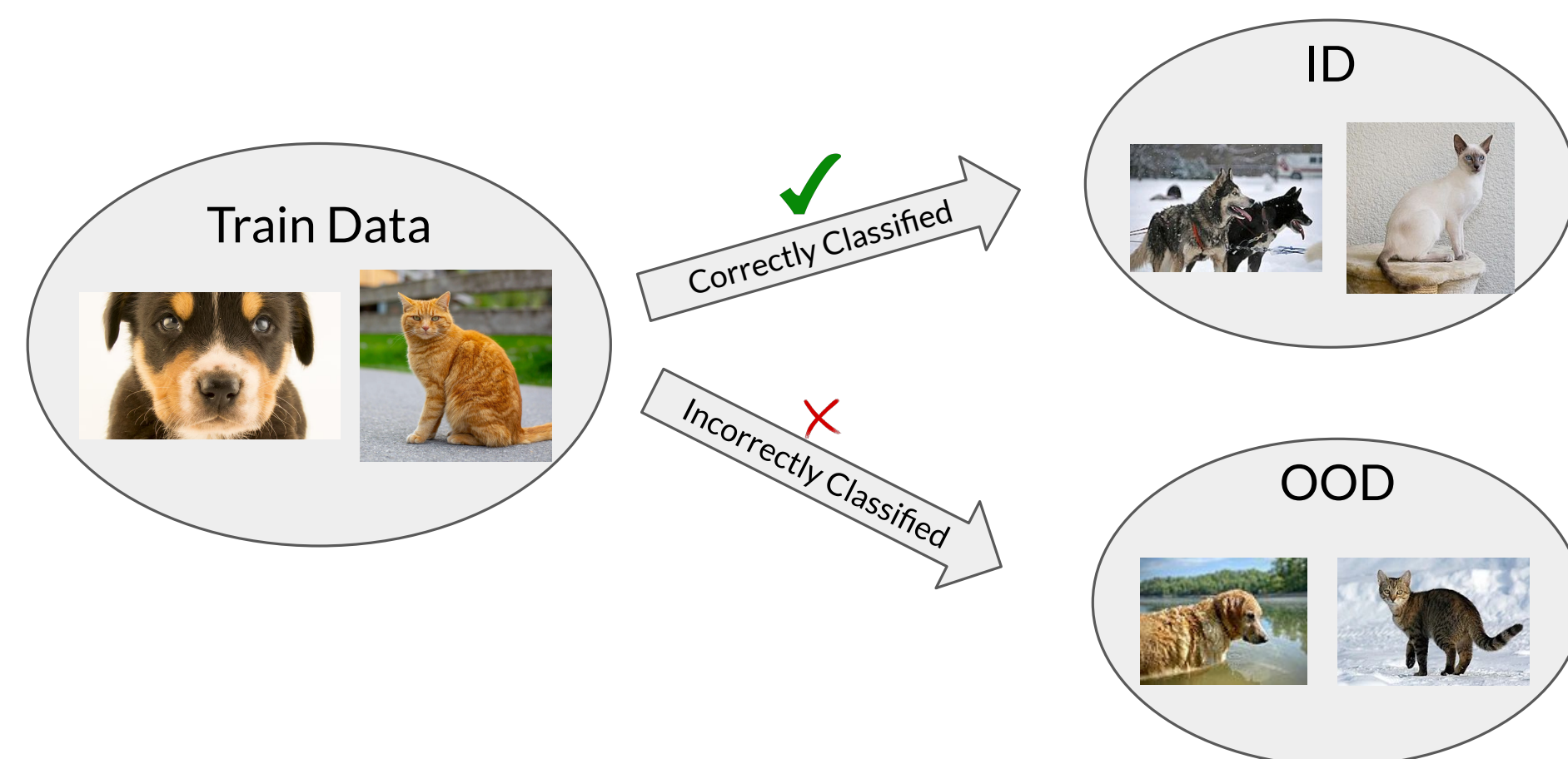


Figure 1. OOD is defined as failure cases in failure-detection.

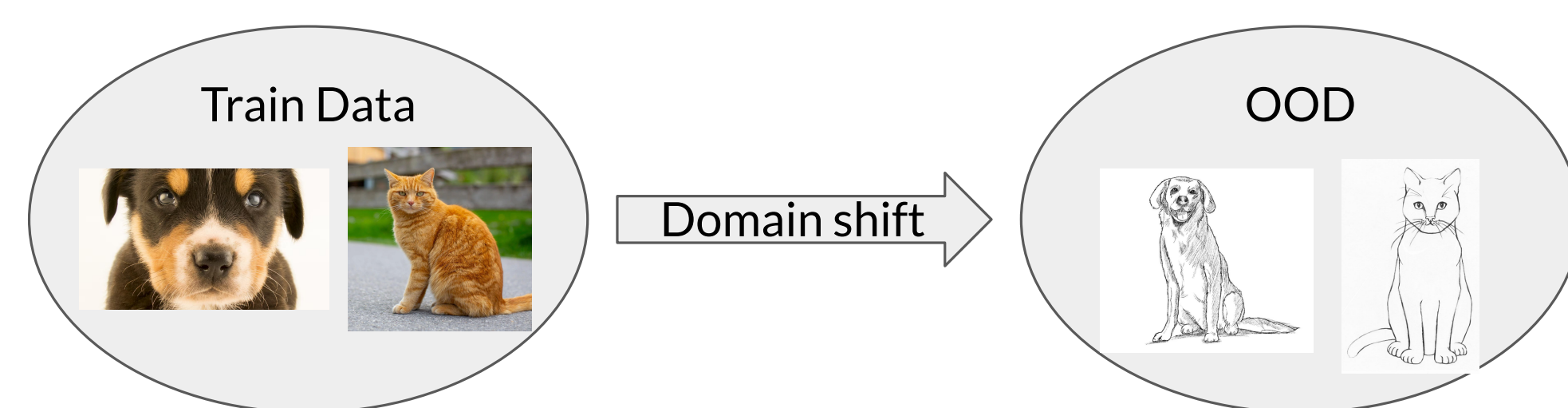


Figure 2. The input style changes but labels are same in Domain Shift detection.

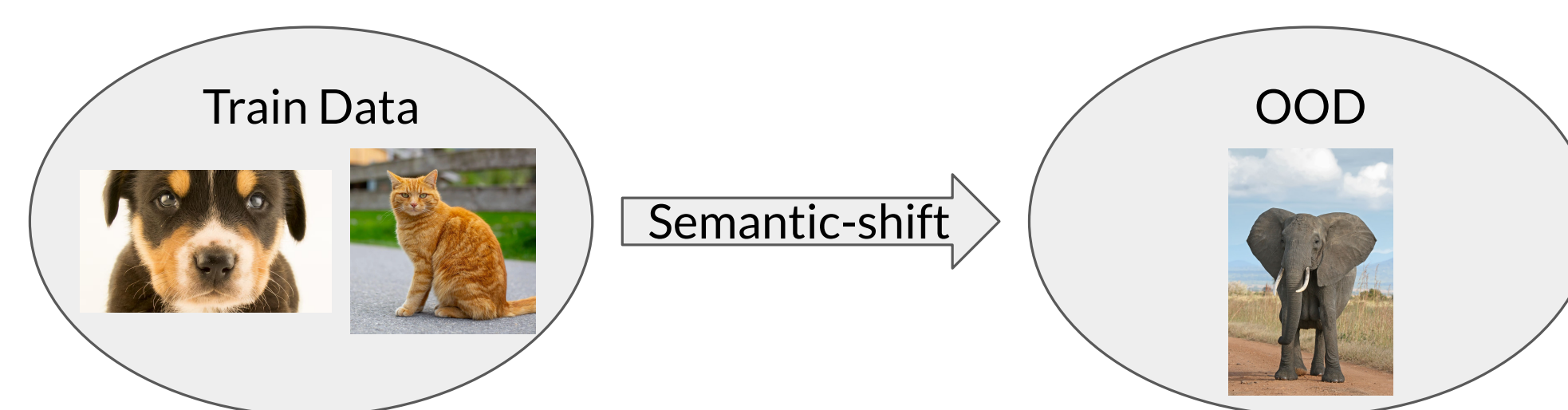


Figure 3. Cases that don't fit into a training label is OOD in new-label detection.

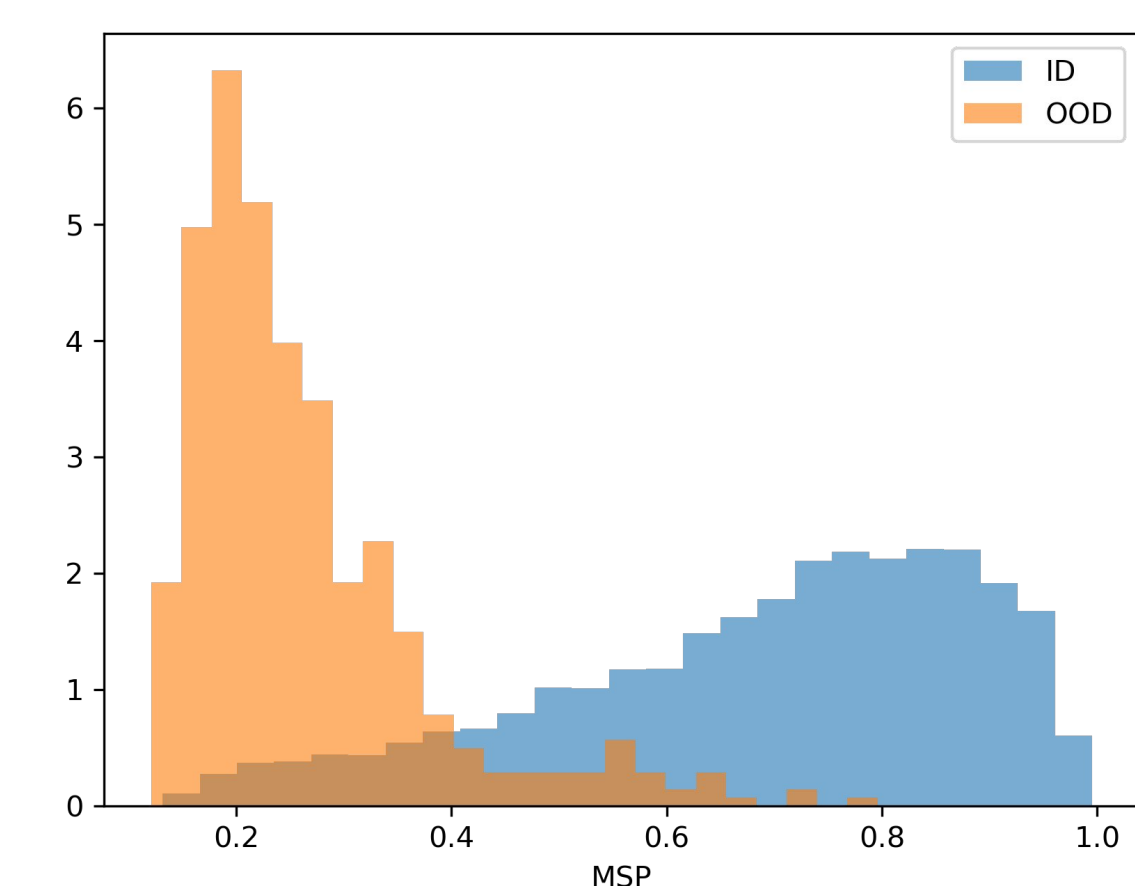


Figure 4. The ID cases are assigned a higher “confidence” by MSP. The goal of an OOD metric is to maximize the ID (blue) examples’ values and minimize the OOD (orange) examples’ values.

Methodology

ImageNette and ImageWoof’s classes were used as the base training classes. The domain-shifted dataset came from the same training classes of ImageNet-Sketch. The new-class labels were a manually selected subset of ImageNet labels (Figure 5).



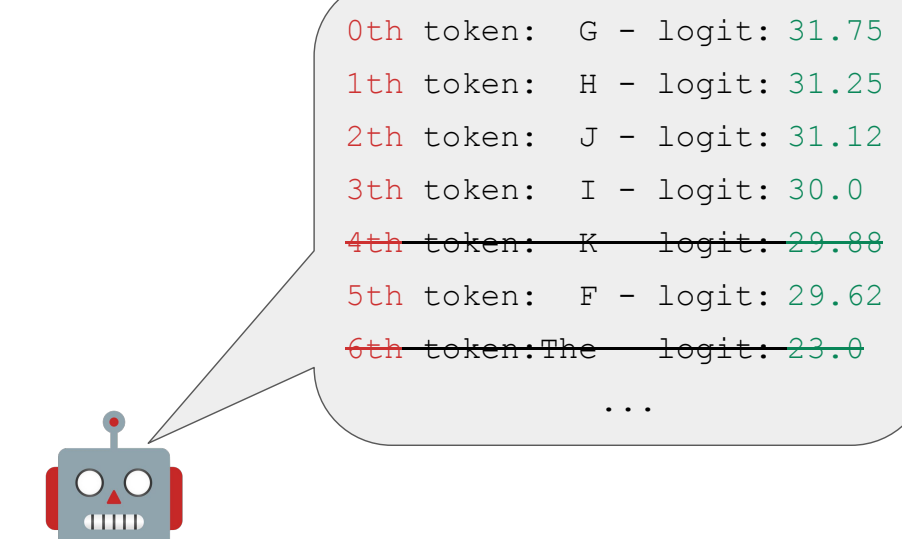
Figure 5. Row 1, 2, and 3 depict examples from ImageNette, ImageWoof, and ImageNet-Sketch, respectively.

The AUROCs of four OOD detection methods were evaluated on failure detection on the training and a new domain, domain-shift detection, and new-class detection on the training and a new domain (Figure 6, 7).



Figure 6. To simulate OOD with VLMs, context learning with randomized labels from A to J for each object was used. VLMs were given 3 examples per label.

Figure 7. Invalid tokens (not A-J) are discarded, though they had low softmax values and tended to have little effect on OOD metrics.



Results

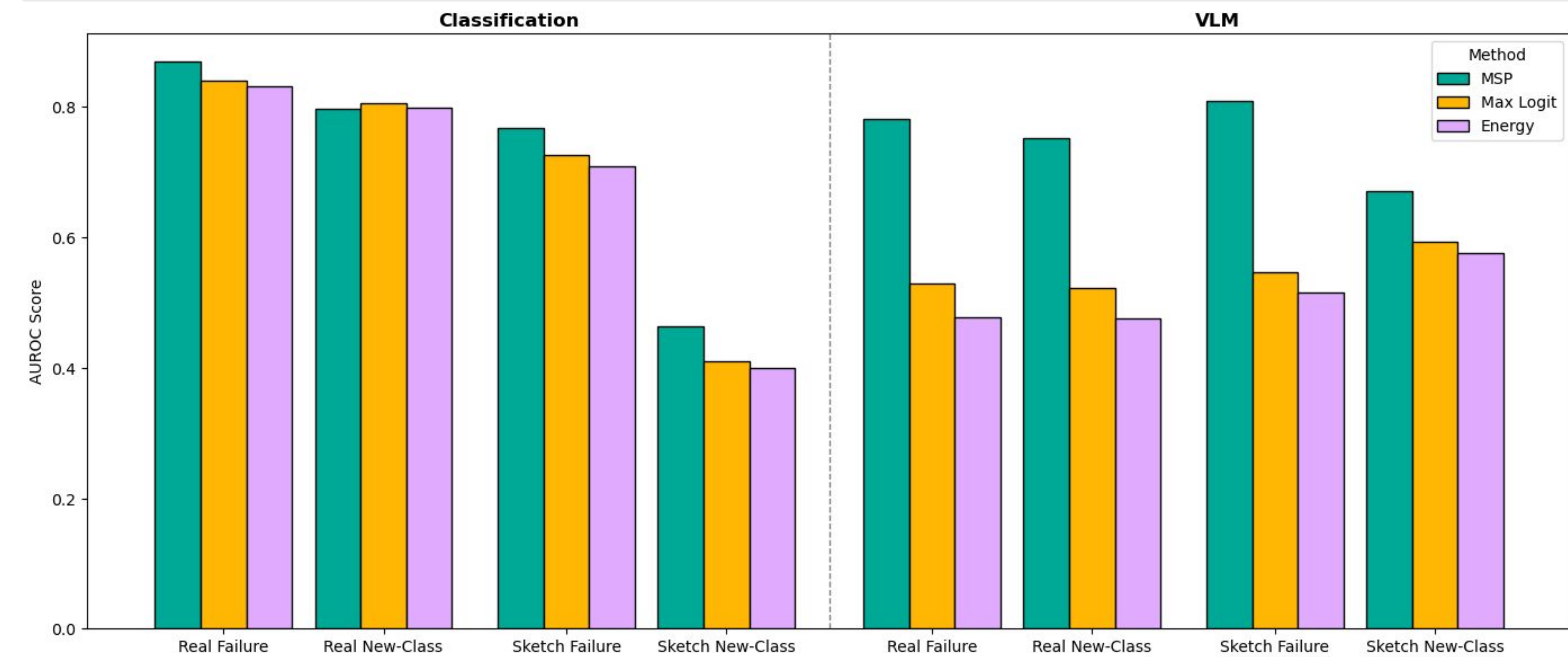


Figure 8. VLMs display some consistent trends on OODs as classification models. For both, MSP is best or matches the best OOD method. However, logit-based methods do not seem to work well on VLMs. VLMs also exhibit the ability to discern new-class on domain-shifted data.

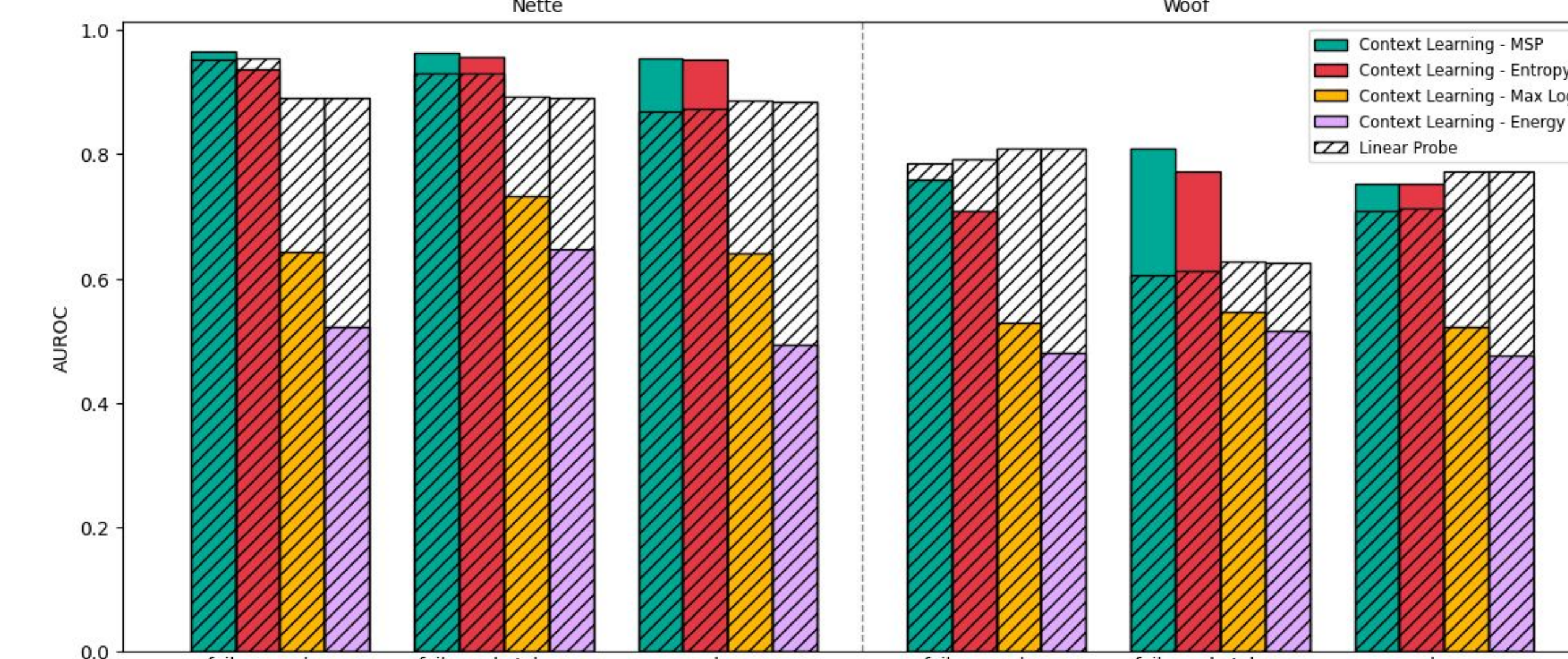


Figure 9. A simple linear probe was trained on the VLM’s visual model with the same dataset. The linear probe generally outperforms the VLM with logit-based methods (max logit and energy), suggesting the language modality of VLMs might have a “normalizing” effect on logits. However, VLM did better or matched the linear probe on softmax-based methods (MSP and Entropy), suggesting the language modality of VLMs still adds nuance.

Analysis of OOD in VLMs reveal:

- MSP still outperforms or matches other OOD metrics for failure and new-class detection in VLMs, similar to narrow classification models.
- Unlike classification models, VLMs retain the ability to detect semantic shift after applying a domain shift (Figure 8).
- The language modality of VLMs have a strong influence on visual classification and OOD detection performance (Figure 9).
- Poor classification performance leads to poor OOD performance (Figure 10, 11).

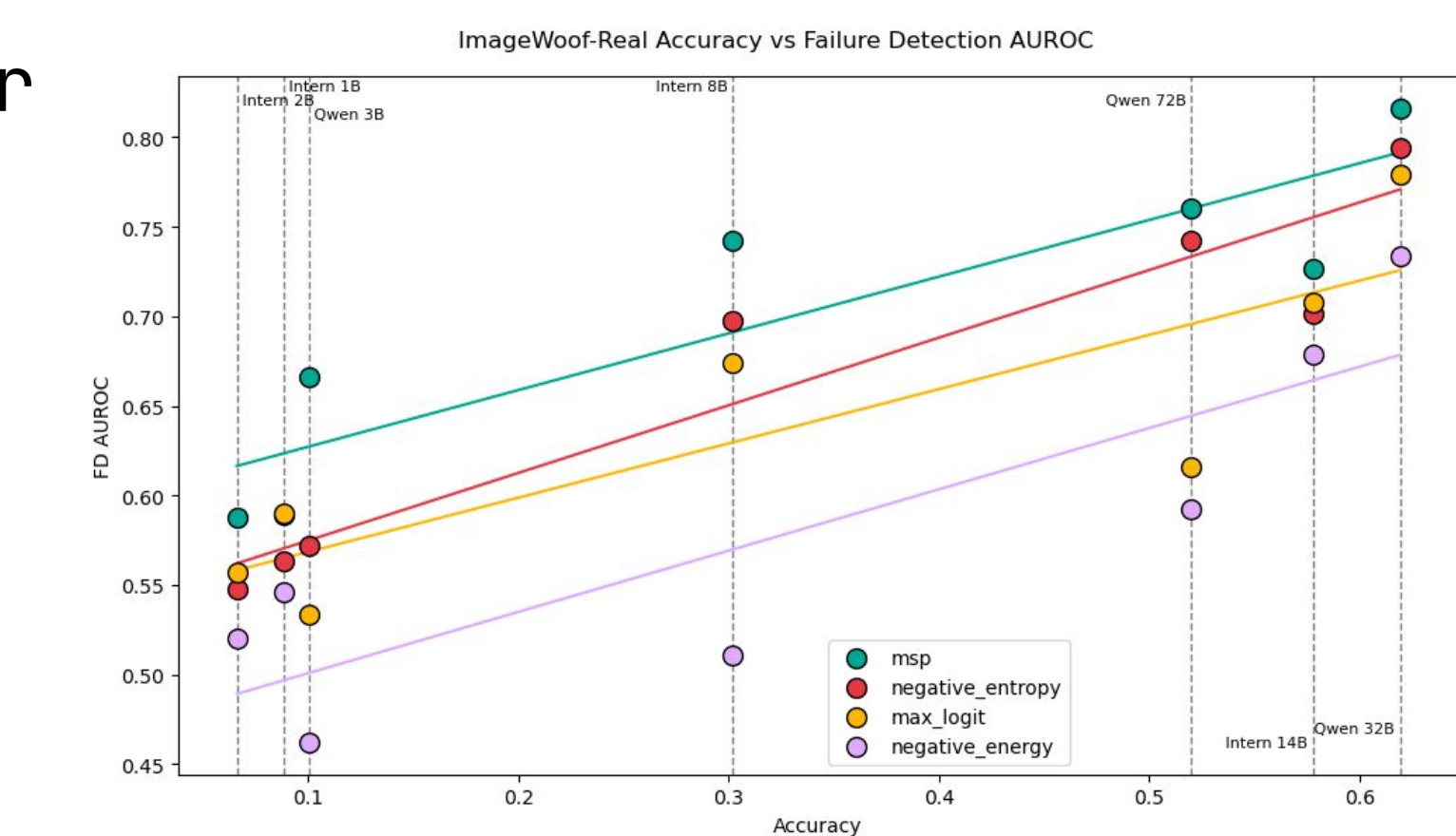


Figure 10. While most models with low accuracy get low AUROC, there are some models with high accuracy and low AUROC.

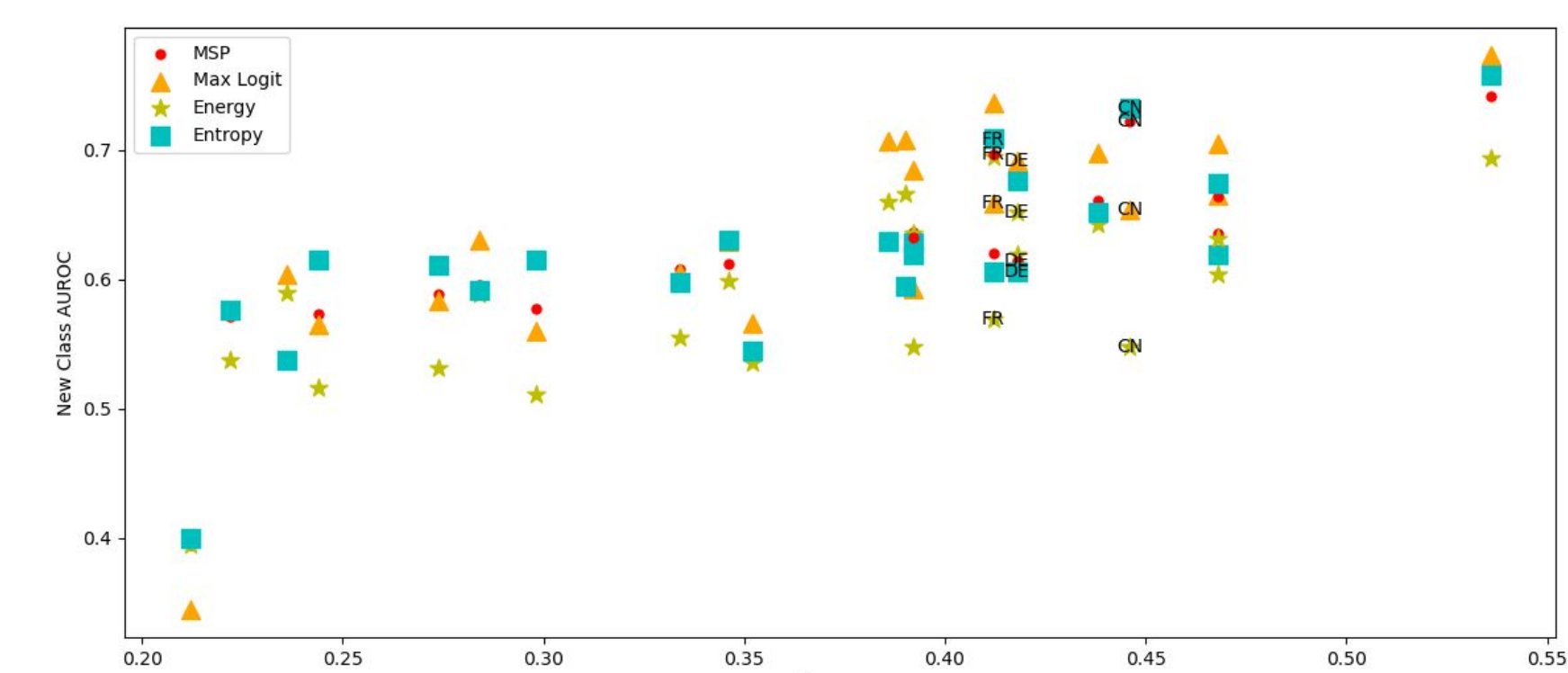


Figure 11. Different paraphrasings of the same prompt causes a high variance in both accuracy and AUROC.

References & Acknowledgements

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